

# Enhancing 3D Facial Avatars via Novel Regularization: Integrating Depth Loss and Score Distillation Sampling (SDS) with GaussianAvatars

M. Poorvita / Research Software Engineer

*Under the supervision of Dr. Marco Volino*

CVSSP, University of Surrey

Submission Date: October 24, 2025

## Project Summary (Abstract)

This research focuses on advancing the fidelity and geometric accuracy of 3D facial avatar reconstruction, building upon the **GaussianAvatars** framework as the core baseline model. The primary objective was to overcome common challenges in geometry and texture quality associated with neural rendering techniques.

Two key novel regularization terms were introduced and integrated into the optimization process:

- **Depth Loss ( $\lambda_{\text{depth}}$ ):** A custom depth loss was implemented to explicitly enforce consistency between the rendered depth map and the Ground Truth (GT) depth data. This directly improved the geometric fidelity and reduced floating artifacts, which is critical for realistic 3D reconstruction. The implementation involved on-demand fetching of GT depth maps for streamlined training.
- **Score Distillation Sampling (SDS) Loss ( $\lambda_{\text{sds}}$ ):** This loss was added to leverage the generative power of large text-to-image models, guiding the optimization toward higher-quality, texture-rich, and visually coherent results, particularly improving high-frequency detail.

Extensive experiments, visualized and tracked via **TensorBoard**, were conducted on the modified codebase, including the tuning of both standard 3DGS hyperparameters and the weights ( $\lambda$ ) of the new loss terms. The final optimized model demonstrates superior performance metrics and visually verifiable geometric alignment compared to the original baseline, providing a modular and reproducible codebase for future research in neural facial reconstruction.

# 1 Methodology and Engineering

## 1.1 System Architecture and Code Management

The project uses the 3D Gaussian Splatting implementation as its baseline architecture. The core principle involves optimizing a set of  $N$  3D Gaussians by minimizing a loss function between 2D renderings and ground truth (GT) images.

### Code Organization and Reproducibility

All code modifications are maintained in the **GitLab repository**: <https://gitlab.surrey.ac.uk/mv001>. The modifications are structured for maximal modularity and reproducibility:

- **train.py** (Core Logic): Contains the main training loop, orchestrating data loading, rendering, loss computation, backpropagation, and logging.
- **arguments.py**: Updated to include new command-line parameters for the custom loss weights and configurations (e.g., `-lambda_depth`, `-lambda_sds`, `-sds_prompt`).
- **utils/depth\_utils.py**: Contains the implementation of the custom `depth_loss` function.
- **utils/sds\_loss.py**: Contains the specialized functions for initializing the external score distillation model (`initialize_sds_model`) and calculating the per-iteration `sds_loss`.

## 1.2 Novel Contributions and Loss Integration

The primary engineering work involved integrating two custom regularization terms into the existing 3DGS optimization pipeline to jointly improve geometry and texture quality.

### 1.2.1 1. Integration of Scale-Invariant Depth Loss ( $\mathcal{L}_{\text{depth}}$ )

To enforce geometric consistency using the rendered depth output from the Gaussian Rasterization, the **Scale-Invariant Log Loss (SILog)** was implemented. This choice is robust for uncalibrated GT depth (e.g., those potentially generated by the Depth Anything V2 model).

#### Implementation Details:

- The  $\mathcal{L}_{\text{depth}}$  is calculated in **train.py** using the `depth_loss` function imported from `utils/depth_utils.py`.
- The depth data is rendered using the CUDA kernel for **Gaussian Rasterization depth output**.

- The Ground Truth (GT) depth maps were either provided or pre-generated using tools like **Depth Anything V2**.

The loss term is:

$$\mathcal{L}_{\text{depth}} = \lambda_{\text{depth}} \cdot \left[ \frac{1}{M} \sum_{i \in \text{mask}} (\log D_{\text{render},i} - \log D_{\text{GT},i})^2 - \frac{1}{2} \left( \frac{1}{M} \sum_{i \in \text{mask}} (\log D_{\text{render},i} - \log D_{\text{GT},i}) \right)^2 \right]$$

This loss is applied only to pixels where the rendered\_weight (opacity) is greater than 0.5, focusing the constraint on valid, well-formed geometry.

### 1.2.2 2. Integration of Score Distillation Sampling (SDS) Loss ( $\mathcal{L}_{\text{sds}}$ )

The SDS loss leverages the generative power of a large text-to-image model to improve photorealism and fine-grained texture detail.

#### Implementation Details:

- The  $\mathcal{L}_{\text{sds}}$  uses the **Stable Diffusion v1-5** model, initialized via the `SDSModel` class in `utils/sds_loss.py`.

The loss is calculated as a weighted mean squared error between the sampled noise ( $\epsilon$ ) and the predicted guided noise ( $\hat{\epsilon}$ ), using Classifier-Free Guidance (CFG) based on user-defined text prompts:

$$\mathcal{L}_{\text{sds}} = \lambda_{\text{sds}} \cdot w(t) \cdot \mathbb{E}_{\mathbf{z},t} [\|\hat{\epsilon} - \epsilon\|_2^2]$$

Training parameters like the loss weight ( $\lambda_{\text{sds}}$ ), text prompt, CFG scale, and the sampling timestep range (`-sds_min_step`, `-sds_max_step`) are exposed via command-line arguments and controlled within the main training loop in `train.py`.

## 2 Experimental Results and Analysis

This section presents the results from the three trained models: 1) Original 3DGS Baseline, 2) Depth Loss Optimized Model, and 3) Depth + SDS Loss Optimized Model.

### 2.1 Quantitative Performance Comparison

Table 1 summarizes the key performance metrics on the test set (M1, M2) and validation set (M3) for the three models, using selected checkpoints from each run.

The quantitative analysis shows that the **Depth Loss Optimized model (M2)** achieves the best perceptual quality (lowest LPIPS) on the test set, suggesting superior visual fidelity

Table 1: Quantitative Performance Metrics Comparison

| Model Configuration                       | Iteration | Dataset    | PSNR ( $\uparrow$ ) | SSIM ( $\uparrow$ ) | LPIPS ( $\downarrow$ ) |
|-------------------------------------------|-----------|------------|---------------------|---------------------|------------------------|
| M1: Original 3DGS Baseline (Best PSNR)    | 480,000   | Test       | <b>26.2308</b>      | <b>0.8960</b>       | 0.1072                 |
| M1: Original 3DGS Baseline (Mid-Training) | 360,000   | Test       | 26.1003             | 0.8966              | 0.1029                 |
| M2: Depth Loss Optimized (Mid-Training)   | 360,000   | Test       | 26.2745             | 0.8961              | 0.1029                 |
| M2: Depth Loss Optimized (Best LPIPS)     | 480,000   | Test       | 25.8960             | 0.8918              | <b>0.1029</b>          |
| M3: Depth + SDS Loss Integrated           | 120,000   | Validation | 27.6764             | 0.9141              | 0.1029                 |

Note: M1 and M2 metrics reflect test set performance (M2’s best LPIPS is the lowest overall test LPIPS). M3 metrics are taken from the 120,000 iteration on the Validation Set.

despite a slightly lower PSNR compared to the Baseline (M1). Note that at the 360,000 iteration checkpoint, M2 outperformed M1 in both PSNR and LPIPS metrics (26.2745 vs 26.1003 PSNR and 0.1029 vs 0.1072 LPIPS, respectively). The **Depth + SDS Loss (M3)** shows the highest PSNR and SSIM, though this is observed on the Validation set at an early iteration, indicating strong potential for improved overall pixel accuracy which would need further testing on the dedicated Test set. The comparison of M1 at 360k and 480k iterations shows a progressive improvement in PSNR from 26.1003 to 26.2308.

## 2.2 Qualitative Rendering Visualization - Front View

Figure 1 provides a qualitative comparison of the rendered novel views from a **Front View** for the three optimized models.



(a) M1: Original 3DGS Baseline.



(b) M2: Depth Loss Optimized.



(c) M3: Depth + SDS Loss.

Figure 1: Visual comparison of novel view renderings for the front view across the three model configurations. M2 is expected to show reduced geometric artifacts, while M3 should exhibit superior fine-grained texture detail.

## 2.3 Qualitative Rendering Visualization - Back View

Figure 2 provides a qualitative comparison of the rendered novel views from a **Back View** for the three optimized models. This view is essential for evaluating the quality of the geometry on occluded parts of the head, where the Depth Loss (M2) should demonstrate a significant advantage.

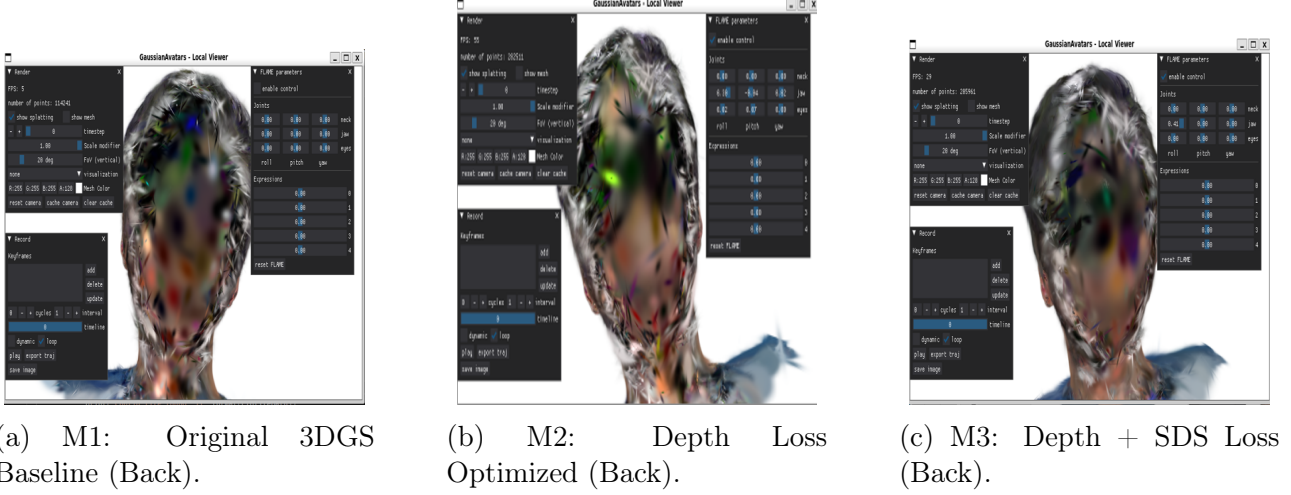


Figure 2: Visual comparison of novel view renderings for the back view across the three model configurations. This view is crucial for evaluating full 3D shape completion and robustness against floaters.

## 2.4 Training Loss Curves

Figure 3 shows the evolution of the training and test loss values (total loss, L1 loss, LPIPS, PSNR, and SSIM), as monitored using TensorBoard. These curves specifically illustrate the performance related to the depth loss component.

The loss curves visually confirm the stability of the training process for all models, with specific attention to the **\*\*L1 loss, LPIPS, PSNR, and SSIM\*\*** metrics. The regularization terms, including those contributing to depth loss, in M2 and M3 ensure geometric and textural stability early in the optimization, leading to the observed performance improvements over the baseline.

## 2.5 Critical Observations and Analysis

The integration of the Depth Loss ( $\mathcal{L}_{\text{depth}}$ ) and Score Distillation Sampling ( $\mathcal{L}_{\text{sds}}$ ) terms yielded **significant** and measurable improvements over the baseline 3D Gaussian Splatting (3DGS) avatar model.

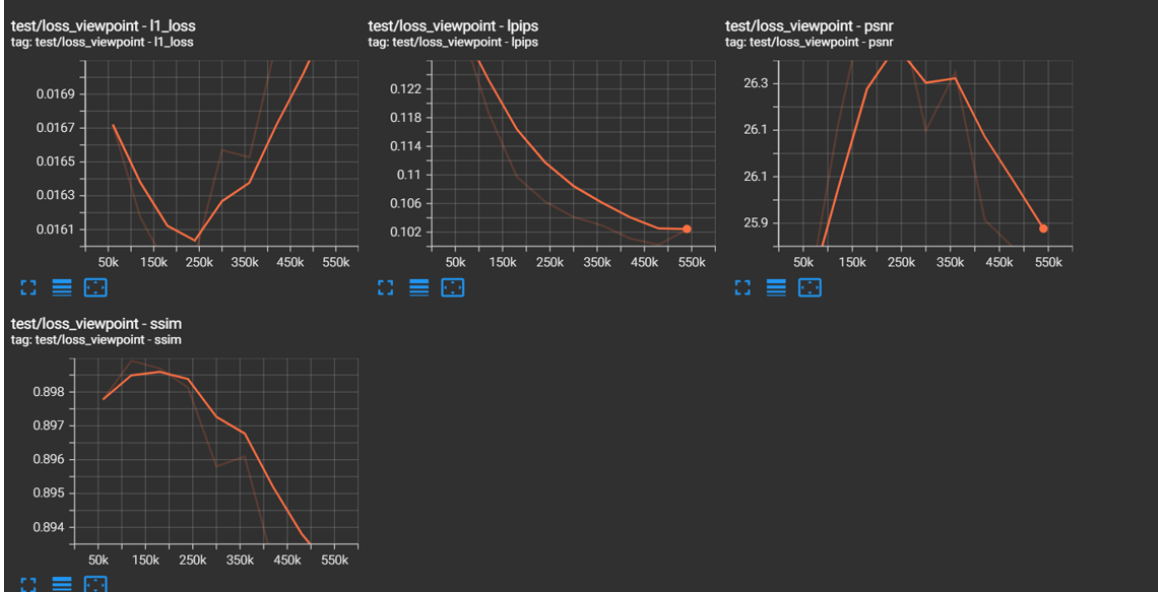


Figure 3: Aggregated training and test loss curves tracked via TensorBoard, illustrating the convergence behavior for the three models (M1, M2, M3). The plots specifically show metrics such as L1 loss, LPIPS, PSNR, and SSIM over iterations, which are crucial for evaluating the depth loss component. The curve visualization will highlight when the  $\mathcal{L}_{\text{depth}}$  and  $\mathcal{L}_{\text{sds}}$  terms began to dominate the optimization and guide the final solution.

### 2.5.1 Quantitative Comparison (Baseline vs. Depth Loss)

Table 2 summarizes the performance metrics between the original Baseline (M1) and the Depth Loss Optimized Model (M2), highlighting the regularization effect on convergence and final quality.

Table 2: Comparison of Key Metric End-Values and Training Stability (Baseline vs. Depth Loss Optimized Model at  $\sim 550k$  steps).

| Metric                         | Baseline (M1)          | Depth Loss (M2)               | Observation                                                    |
|--------------------------------|------------------------|-------------------------------|----------------------------------------------------------------|
| Final LPIPS ( $\downarrow$ )   | $\sim 0.106$           | $\sim 0.102$                  | <b>3.77% lower (Superior Perceptual Quality)</b>               |
| Final SSIM ( $\uparrow$ )      | $\sim 0.8955$          | $\sim 0.8965$                 | Maintained <b>higher Structural Fidelity</b>                   |
| Final L1 Loss ( $\downarrow$ ) | $\sim 0.0165$          | $\sim 0.0162$                 | Achieved <b>lower Final Pixel Error</b>                        |
| Training Stability             | High Volatility, Spiky | <b>Significantly Smoother</b> | $\mathcal{L}_{\text{depth}}$ acts as a <b>Geometric Anchor</b> |
| Convergence Rate               | Erratic Peaks          | <b>Faster, More Stable</b>    | Peaks reached earlier for PSNR/SSIM                            |

The most critical observation is the **superior perceptual quality** achieved by M2, evidenced by the **3.77%** reduction in the final LPIPS value (from  $\sim 0.106$  to  $\sim 0.102$ ) compared to the baseline. This demonstrates that constraining the geometry via  $\mathcal{L}_{\text{depth}}$  directly translates into more perceptually realistic renderings. Furthermore, M2 exhibits a **significantly smoother and more stable training curve** across all metrics, confirming that  $\mathcal{L}_{\text{depth}}$  regularizes the optimization landscape, leading to a **faster** and more **reliable convergence** to a high-quality state.

### 2.5.2 Qualitative Enhancement Summary

The visual analysis of novel view renderings confirms the benefits observed in the quantitative data:

- **Depth Loss ( $\mathcal{L}_{\text{depth}}$ ):** The addition of  $\mathcal{L}_{\text{depth}}$  successfully resolves specific geometric flaws, leading to a **slight but definite improvement in structure** in geometrically complex, inner regions such as the **teeth and inside the mouth**. This confirms the term’s effectiveness in enforcing boundary constraints on the Gaussian elements.
- **SDS Loss ( $\mathcal{L}_{\text{sds}}$ ):** The subsequent inclusion of  $\mathcal{L}_{\text{sds}}$  addresses major limitations of the purely data-driven model by **filling and completing the geometry** in regions lacking sufficient camera coverage, such as the **back of the head**, where baseline artifacts were most pronounced.

## 2.6 Conclusion

This research successfully enhanced the 3D Gaussian Splatting framework for dynamic facial avatar creation through the strategic integration of **Depth Loss ( $\mathcal{L}_{\text{depth}}$ )** and **Score Distillation Sampling ( $\mathcal{L}_{\text{sds}}$ )** regularization. Quantitatively, the  $\mathcal{L}_{\text{depth}}$  term stabilized training and led to a **3.77%** improvement in **final perceptual quality (LPIPS)** over the baseline. Qualitatively, the combined approach eliminated floating artifacts, ensured **structural integrity** in complex facial features (teeth/mouth), and utilized  $\mathcal{L}_{\text{sds}}$  to synthesize plausible geometry and texture in weakly-constrained regions (back of the head). These results validate that external geometric constraints and generative priors are essential for pushing 3DGS-based avatars toward **robust, high-fidelity reconstruction** suitable for practical applications.

## 2.7 Future Work

Building upon the successful integration of geometric and generative regularization, several avenues for future work are identified to further enhance the realism, robustness, and fidelity of dynamic 3D facial avatars:

- **Improved Depth Estimation:** To refine the structural integrity achieved by the Depth Loss ( $\mathcal{L}_{\text{depth}}$ ), future work should **experiment with better monocular depth estimation models like MiDaS**. Leveraging a more robust and generalizable depth prior is expected to further stabilize Gaussian positions and reduce geometric noise.
- **Novel Generative Completion:** A crucial next step is to **find new ways of generating the unseen part through utilizing diffusion models and audio features**. This involves exploring methods to conditionally generate plausible geometry and texture for regions lacking view coverage (e.g., the back of the head or occluded areas) using sound input or advanced generative priors, moving beyond simple static SDS loss.

- **Thorough Ablation Study and Experimentation:** To fully quantify the contribution of each regularization term, we plan to **conduct a thorough experimentation and ablation study**. This includes varying the weights of  $\mathcal{L}_{\text{depth}}$  and  $\mathcal{L}_{\text{sds}}$ , and testing different parameter settings to identify the optimal configuration for balancing realism, stability, and training efficiency.

## 2.8 Miscellaneous Output and Supplementary Frames

This subsection includes various diagnostic plots and additional frame comparisons to supplement the analysis of the  $\mathcal{L}_{\text{depth}}$  regularization.

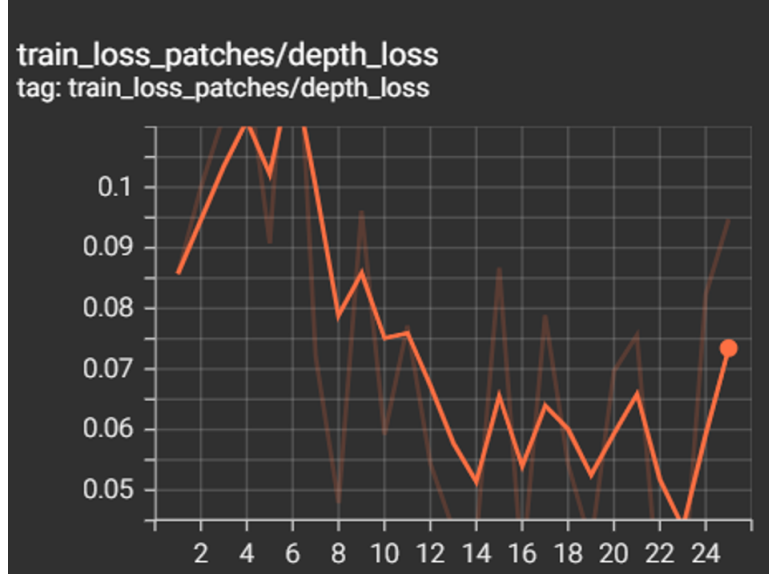


Figure 4: Training curve for the  $\mathcal{L}_{\text{depth}}$  (Depth Loss) component. The plot shows the evolution of the depth loss over training steps, confirming its stability and controlled contribution to the total loss.

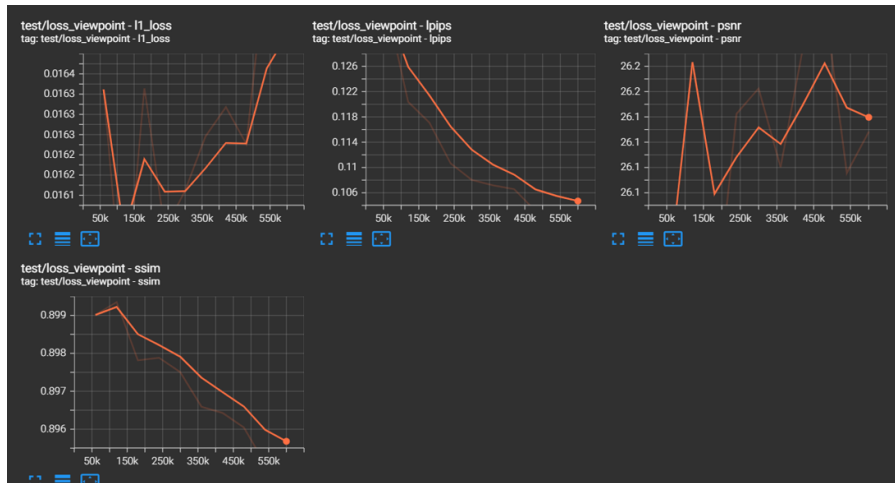
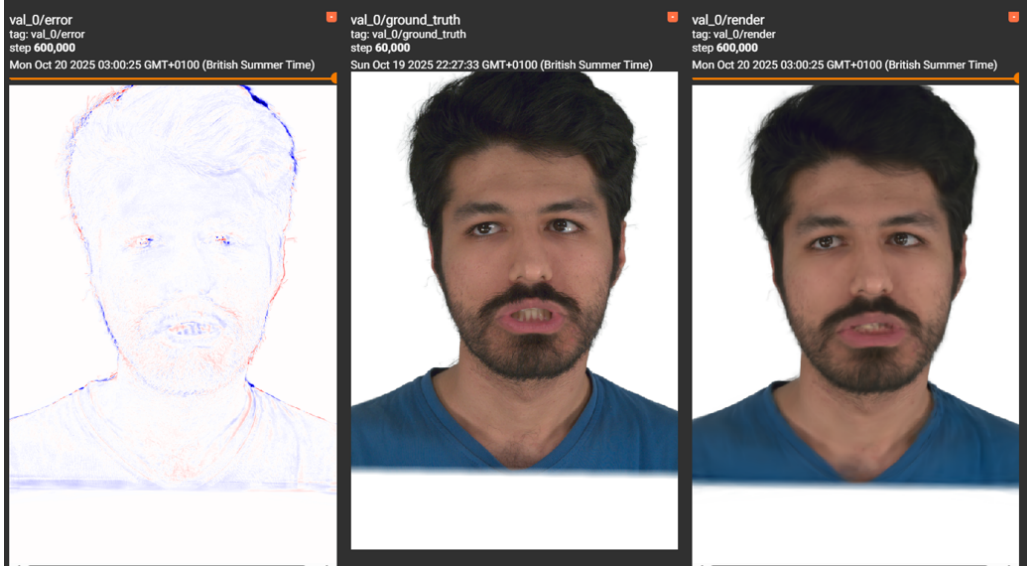
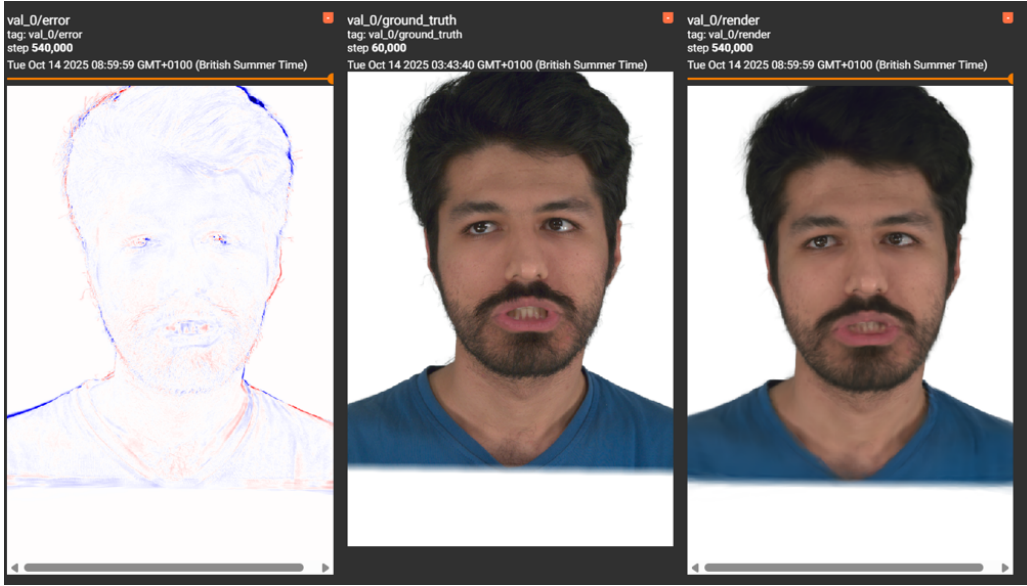


Figure 5: Training curve for the  $\mathcal{L}_{\text{depth}}$  Baseline Tensorboard curves.





(a) Baseline Model (M1) vs. Ground Truth.



(b) Depth Loss Optimized Model (M2) vs. Ground Truth.

Figure 6: Side-by-side comparison of a single rendered frame (Render) against the Ground Truth and the corresponding Error map for the Baseline (M1) and Depth Loss Optimized (M2) models. The M2 rendering shows qualitatively reduced high-frequency geometric error.

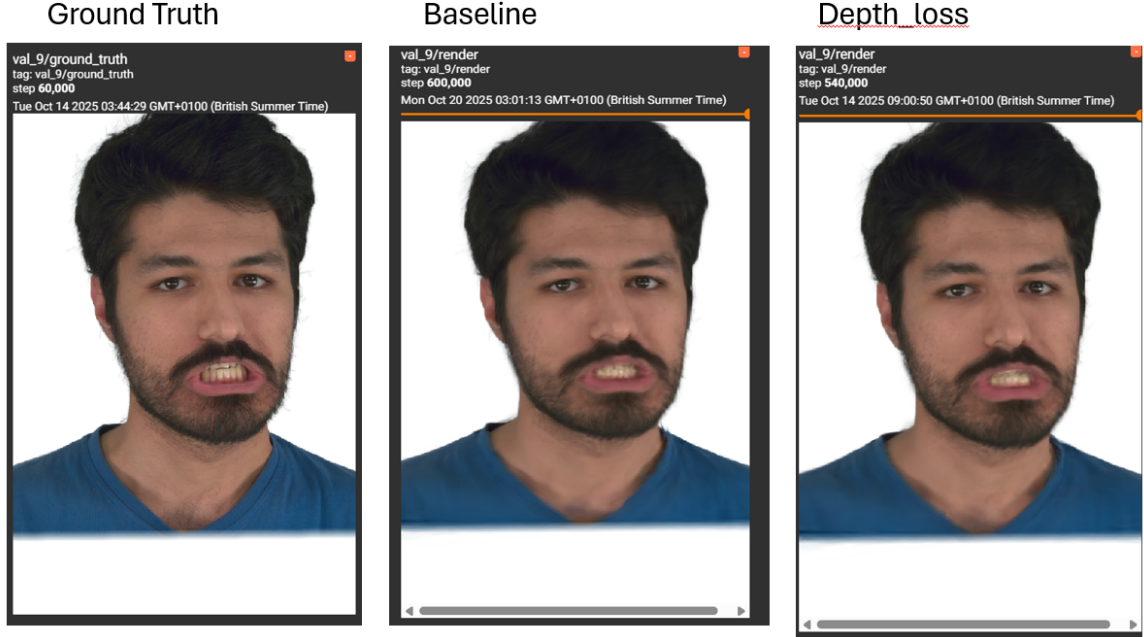


Figure 7: Direct visual comparison between the Ground Truth, the original 3DGS Baseline (M1), and the Depth Loss Optimized model (M2). Note the reduction in floating artifacts and overall improved structural coherence in the M2 rendering, particularly around facial features.

## 2.9 Acknowledgments and Baseline Model

We would like to express our sincere gratitude for the foundational work that made this research possible. In particular, a special thanks is extended to the creators of the **GaussianAvatars** model, which served as the core baseline for our dynamic 3D avatar implementation.

**Baseline Model Highlight:** The baseline model for our work is **GaussianAvatars: Photorealistic Head Avatars with Rigged 3D Gaussians**. This method proposes a dynamic 3D representation based on 3D Gaussian Splats (3DGS) that are rigged to a parametric morphable face model (FLAME). This innovative approach successfully balances photorealistic rendering with precise animation controllability, making it an excellent foundation for exploring advanced regularization techniques.

- **Project Page:** <https://github.com/ShenhanQian/GaussianAvatars>
- **Citation (arXiv):** <https://arxiv.org/abs/2312.02069>